Scripted Data Collection and Exploration

14.33 Research and Communication in Economics

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Novel data as a contribution

Paper	Data purpose	Data novelty
D'Amico, Soltas, and Wang ('26)	Opportunity costs of permitting	US coverage from cities' websites + FOIA requests
Walker and Wooders ('01)	Nash Eq. in tennis	Handcoded match videos
Carlson, Correia, and Luck ('22)	Banking competition and stability	OCR of archived 19th c. bank reports
Cavallo and Rigobon ('16)	Daily prices and inflation	Scraped retailer websites
Baker, Bloom, and Davis ('16)	Economic policy uncertainty	Word counts in newspaper text
Henderson, Storeygard, and Weil ('12)	Economic growth in developing world	Satellite images of night lights

Cheaper collection makes ambitious data feasible

- You exercise judgement in identifying nonobvious data sources
- Agents write scrapers, parsers, and API clients in minutes

Linking existing sources can create a new dataset

- Crosswalks: geographic codes, industry codes, identifiers that change over time
- Fuzzy matching: firm, lender, or school names across sources
- Example: Feigenbaum ('18) matches names in Iowa State Census of 1915 and 1940 Federal Census to estimate intergenerational mobility

Agenda

1. Ways to obtain data
2. Types of data
3. Efficient tabular data management

Ways to obtain data

- Use the most structured route that exists
- Less structured routes produce data that others do not have

Data extraction methods

Route	Example	Main risk
Download	Census, BLS, and IPUMS files	Unrecorded version changes
API	SEC EDGAR, Census API	Keys and rate limits
Web scraping	Public websites	Blocked requests, terms of use
Public-records request	Government files	Delay, denial, fees
Archives	Scanned reports, historical newspapers	OCR errors

General best practices

- **Scripting**:When possible, extract raw data through a script. Test first on data subset.
- **Versioning**: Store unedited raw data and record access date/version

API best practices

- **Security**: Read API keys from environment variables (if using a `.env` file, Git-ignore it), never hard-code them in a tracked script
- **Rate-limiting**: Check the API's limits beforehand and build in sleep times accordingly
- **Size**: Query only the rows and columns you need, and stream if the file would be out-of-memory

Web scraping best practices

- **Check for alternatives**: Look for an API or a bulk download first
- **Transparency**: Identify yourself in a User-Agent header
- **Access rules**: Read the terms of use and `robots.txt`
- **Rate-limiting**: Implement delays between requests to avoid mimicking a Denial-of-Service (DoS) attack
- **Detect rendering**: Target static HTML code, or hidden APIs behind JavaScript-rendered sites

Public-record request best practices

- **Check existing availability**
  - → if data are vended publicly (e.g., NYC Open Data)
  - → if there are existing requests (e.g., MuckRock, Data Liberation Project)
- **Expect delays**
  - → response deadlines of 20 business days for federal, 10 business days for MA
  - → delivery often takes longer
- **Be specific***
  - → Ask for an existing database export in an electronic format
  - → Specify fields, years, and record system

Archives best practices

- **Check availability**: Search library and institutional repositories before scanning yourself
  - → HathiTrust
  - → Internet Archive
  - → Chronicling America
- **Native text availability**: Prefer sources with a text layer rather than image-only scans

Types of data

- Your final analysis will be tabular: columns of variables and rows of observations
- But there are many ways to get there!
- Six nonstandard examples: JSON, PDFs, spatial data, networks, text, images/video/audio

Tabular data is the target format

```
county     year  state  unemp_rate  min_wage
Suffolk    2024  MA     4.1         15.00
Cheshire   2024  NH     2.6          7.25
```

- Some column or combination of columns should uniquely identify observations (“primary key”)
- String categories can be converted to numeric as a set of dummy variables (“one-hot encoding”)

Example 1: JSON

- JSON is a common output format of APIs
- Efficiently stores nested relationships

Raw JSON:

```
{"cik": "0000320193", "name": "Apple Inc.", "sic": "3571",
 "filings": [{"form": "10-Q", "filingDate": "2026-07-31"},
             {"form": "10-Q", "filingDate": "2026-05-01"}]}
```

Target tables:

Table	One row per	Key
`firms`	Firm	`cik`
`filings`	Filing	`cik` and filing number

Example 2: PDFs

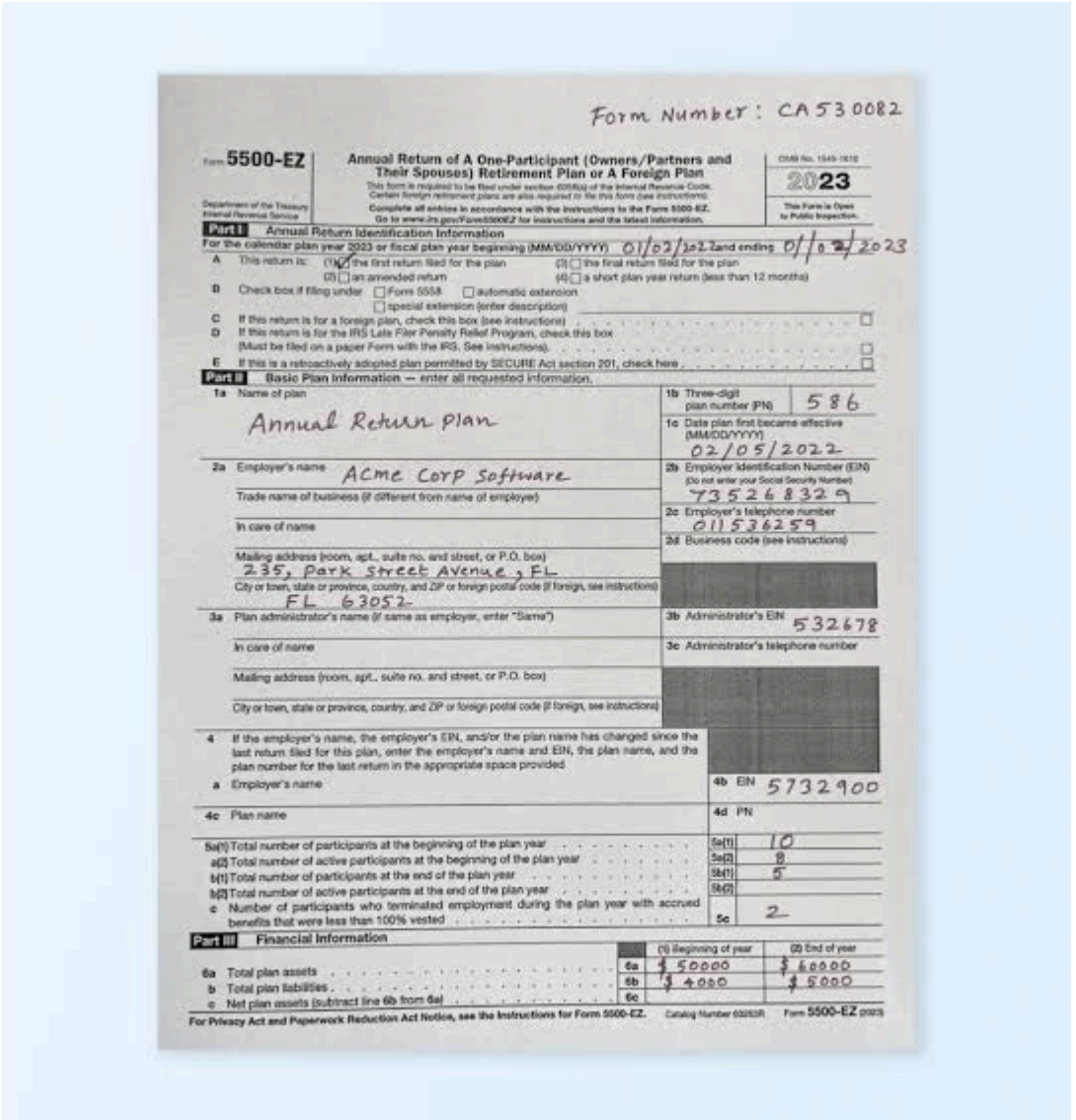
- If it has a native text layer, directly extract it:
  - → **pdfplumber** for text and tables
  - → **Camelot** for tables
- If not, use optical character recognition (OCR):
  - → **Marker** for open-source, CPU/GPU-compatible OCR
  - → **Docling** + **Tesseract** for OCR on CPU
- **OCR leaderboards** show multimodal AI models have state-of-the-art performance

Example 2a: PDF with native text layer

Table A. Job openings, hires, and total separations by industry, seasonally adjusted

Category	Job openings			Hires			Total separations		
	Aug. 2025	July 2026	Aug. 2026[p]	Aug. 2025	July 2026	Aug. 2026[p]	Aug. 2025	July 2026	Aug. 2026[p]
LEVELS BY INDUSTRY (in thousands)									
Total	6,919	7,335	7,079	5,145	5,146	5,192	5,207	5,128	5,070

Example 2b: Structured PDF scan



Form Number: CA530082

Form 5500-EZ

Annual Return of A One-Participant (Owners/Partners and Their Spouses) Retirement Plan or A Foreign Plan

This form is required to be filed under section 6058(a) of the Internal Revenue Code. Certain foreign retirement plans are also required to file this form (see instructions).

Complete all entries in accordance with the instructions to the Form 5500-EZ. Go to www.irs.gov/Form5500EZ for instructions and the latest information.

OMB No. 1545-1810

2023

This Form is Open to Public Inspection.

Department of the Treasury Internal Revenue Service

Part I Annual Return Identification Information

For the calendar plan year 2023 or fiscal plan year beginning (MM/DD/YYYY) 01/02/2022 and ending 01/02/2023

A This return is: (1) ☑ the first return filed for the plan (3) ☐ the final return filed for the plan (2) ☐ an amended return (4) ☐ a short plan year return (less than 12 months)

B Check box if filing under ☐ Form 5558 ☐ automatic extension ☐ special extension (enter description)

C If this return is for a foreign plan, check this box (see instructions) ☐

D If this return is for the IRS Late Filer Penalty Relief Program, check this box (Must be filed on a paper Form with the IRS. See instructions). ☐

E If this is a retroactively adopted plan permitted by SECURE Act section 201, check here ☐

Part II Basic Plan Information — enter all requested information.

1a Name of plan: Annual Return plan

1b Three-digit plan number (PN): 586

1c Date plan first became effective (MM/DD/YYYY): 02/05/2022

2a Employer's name: ACme Corp Software

2b Employer identification Number (EIN) (Do not enter your Social Security Number): 735268329

Trade name of business (if different from name of employer)

2c Employer's telephone number: 011536259

In care of name

2d Business code (see instructions)

Mailing address (room, apt., suite no. and street, or P.O. box): 235, Park Street Avenue, FL

City or town, state or province, country, and ZIP or foreign postal code (if foreign, see instructions): FL 63052-

3a Plan administrator's name (if same as employer, enter "Same")

3b Administrator's EIN: 532678

In care of name

3c Administrator's telephone number

Mailing address (room, apt., suite no. and street, or P.O. box)

City or town, state or province, country, and ZIP or foreign postal code (if foreign, see instructions)

4 If the employer's name, the employer's EIN, and/or the plan name has changed since the last return filed for this plan, enter the employer's name and EIN, the plan name, and the plan number for the last return in the appropriate space provided

a Employer's name

4b EIN: 5732900

4c Plan name

4d PN

5a(1) Total number of participants at the beginning of the plan year	5a(1)	10
a(2) Total number of active participants at the beginning of the plan year	5a(2)	8
b(1) Total number of participants at the end of the plan year	5b(1)	5
b(2) Total number of active participants at the end of the plan year	5b(2)	
c Number of participants who terminated employment during the plan year with accrued benefits that were less than 100% vested	5c	2

Part III Financial Information

		(1) Beginning of year	(2) End of year
6a Total plan assets	6a	$ 50000	$ 60000
b Total plan liabilities	6b	$ 4000	$ 5000
c Net plan assets (subtract line 6b from 6a)	6c		

For Privacy Act and Paperwork Reduction Act Notice, see the Instructions for Form 5500-EZ. Catalog Number 63263R Form 5500-EZ (2023)

Example 2c: Unstructured PDF scan



Store #05666
3515 DEL MAR HTS,RD
SAN DIEGO, CA 92130
(858) 792-7040

Register #4 Transaction #571140
Cashier #56661020 8/20/17 5:45PM

wellness+ with Plenti
Plenti Card#: 31XXXXXXXXXX4553
1 G2 RETRACT BOLD BLK 2PK 1.99 T
SALE 1/1.99, Reg 1/4.69
Discount 2.70-

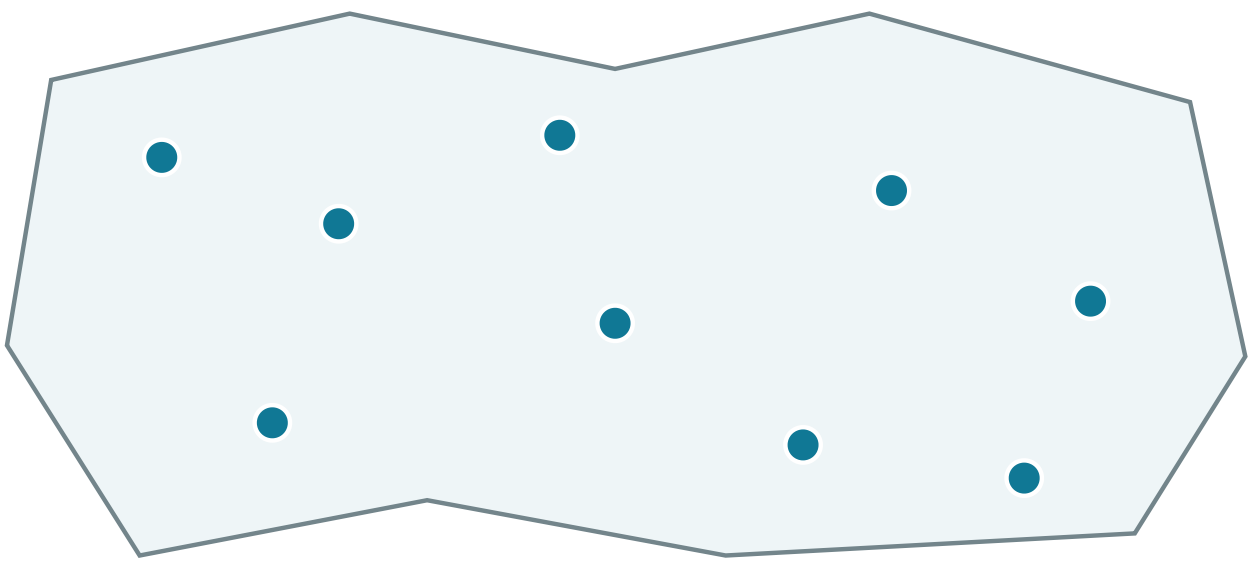
1 Items Subtotal 1.99
Tax .15
Total 2.14
MASTER 2.14
MASTER card * #XXXXXXXXXXXX5485
App #AA APPROVAL AUTO
Ref # 05639E
Entry Method: Chip

Example 3: Spatial

- Put all layers in the same coordinate reference system before joining
- Transform spatial data into a table, such as distance from a fixed location for each unit

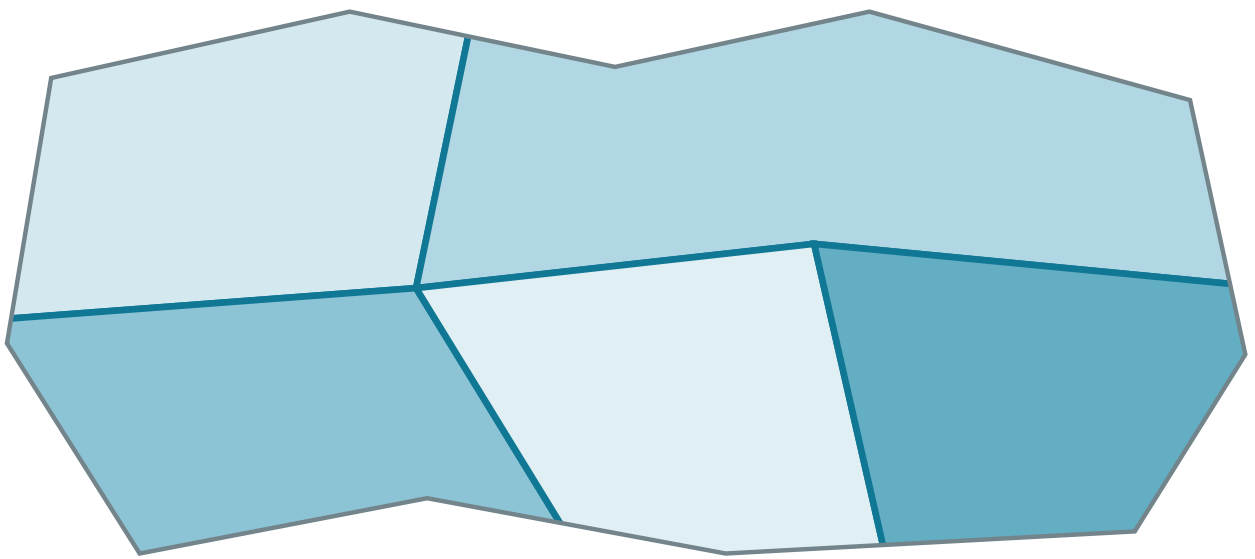
Example 3a: Spatial, vector points

- Vector points represent individual locations, such as bank branches



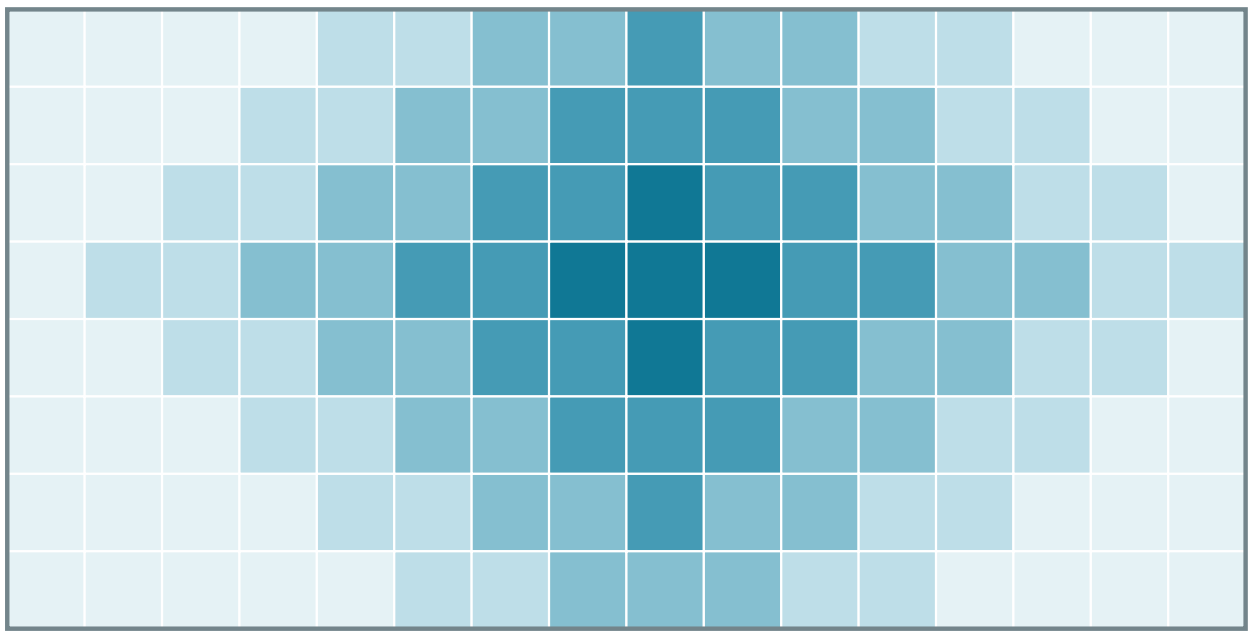
Example 3b: Spatial, vector polygons

- Vector polygons represent areas, such as counties

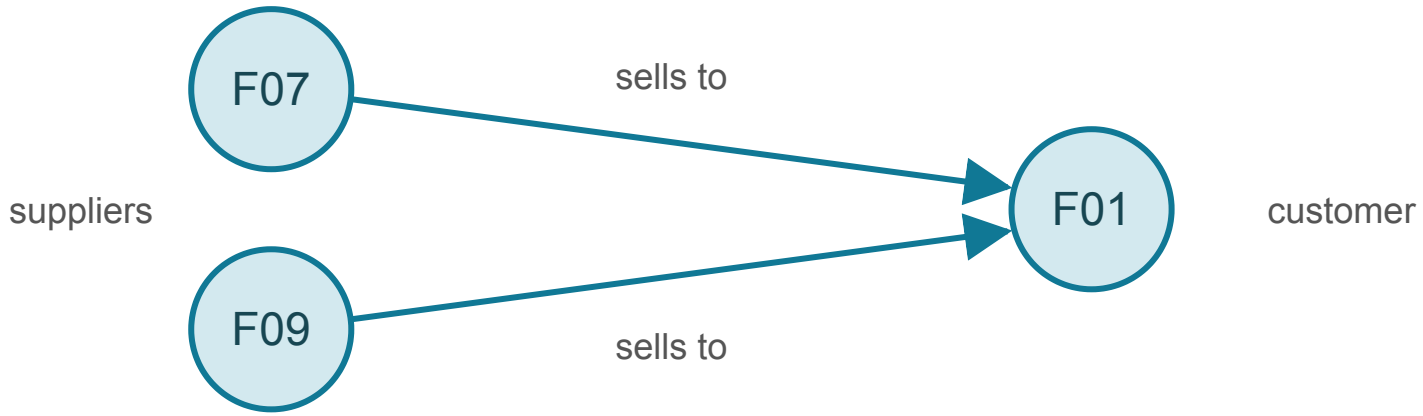


Example 3c: Spatial, raster

- A raster stores one value per grid cell, such as population density



Example 4: Networks



Target tables: one row per node and one row per edge

```
nodes:  firm_id  industry        edges:  supplier  customer  year  sales
        F01      auto                    F07       F01       2004  ...
        F07      parts                   F09       F01       2004  ...
        F09      parts
```

- Examples: friend connections, supply chains, financial intermediaries
- Survey: Graham (2020), "Network Data," *Handbook of Econometrics*

Example 5: Text

```
doc_id  firm  year  paragraph                          tariff_risk
0412    DE    2025  "Changes in trade policy could..."  1
0413    DE    2025  "Our dealers depend on credit..."   0
```

- Measure with keyword counts, a trained classifier, topics, or a language model
- Non-LLM resource: Gentzkow, Kelly, and Taddy (2019), “Text as Data”
- LLM resource: Expected Parrot

Example 6: Images, Audio, or Video

- Multimodal LLMs can transform images, audio, or video (and, of course, text!) to structured numeric output
  - → Image example: Ludwig and Mullainathan ’24 on mugshots predicting judge decisions
  - → Audio example: Gorodnichenko et al ’23 on emotion in Federal Reserve chair speech
  - → Video example: Dietrich ’21 on politicians literally crossing the aisle on C-SPAN
- Using a frontier model through an API generally incurs usage costs
- Open-weight models can run locally through a provider like Ollama depending on size, especially on the MIT server’s GPUs
- In any case, validate a random sample with human annotation or source-of-truth measurements (Dell and Rambachan ’26)

Efficient tabular data management

- Script the workflow and save definitions and checks
- Store large tables in Parquet, keep datasets out of Git
- Keep local data backed up to Dropbox
- Use DuckDB to select, summarize, and join before loading results

Script the workflow once and reuse results

Raw downloads → cleaned data → analysis output → tables and figures

- Keep downloads unchanged, with source URLs, dates, and versions
- Script cleaning, joins, and descriptive statistics and analysis

Record lessons learned for yourself and agents

- **Schema**: codebook with meanings, types, units, codes, and year coverage
- **Decisions**: sample rules, recodes, and crosswalks in project files
- **Verification**: test an appropriate random sample, check counts/keys/missingness
- **Exploration**: distributions, trends, and correlations; figures for you and CSVs for agents

Opinionated efficient tabular data storage: Parquet

- Apache Parquet is a highly compressible columnar file type for tables
- Tradeoff is that there is no simple spreadsheet double-click inspection

  → Can inspect raw data in an interactive session when loaded locally, or export extract to a CSV

Mortgage data	CSV	Parquet	Smaller by
2007–2024	70 GB	6 GB	About 12×

Keep large datasets local and backed up

- **Never commit large datasets to Git**, including derived tables
- Add data paths to `.gitignore` before the first download
- Keep source data available locally and backed up in the cloud
- Commit scripts, codebooks, and small crosswalks

Example `.gitignore`:

```
/data/raw/
/data/clean/
*.parquet
```

Opinionated efficient tabular data I/O: DuckDB

DuckDB is a local database engine based on SQL syntax

- **Flexible**: query Parquet, CSV, and JSON files directly
- **Selective**: return only the needed rows, columns, aggregates, or joined results
- **Out-of-memory capabilities**: can use temporary disk space when data exceed RAM
- **Fast**: optimized for filtering, grouped summaries, and joins

100 MILLION ROWS, 5 GB, FIVE GROUPED SUMMARIES

DuckDB	Python `pandas`	R `dplyr`
2 seconds	42 seconds	error

Using DuckDB

Multiple ways to apply the same SQL query logic

```sql
SELECT county, year, COUNT(*) AS originations
FROM 'data/derived/hmda/*.parquet'
WHERE action = 1
GROUP BY county, year;
```

Interface	Run the saved query	Purpose
Python package	`duckdb.sql(query).df()`	Reproducible scripting
R package through DBI	`dbGetQuery(con, query)`	Reproducible scripting
Command-line interface (CLI)	Start `duckdb`, enter SQL	Native terminal experience
Local browser UI	Start `duckdb -ui`	Interactive exploration

Summary: from novel sources to usable data

- **Find a contribution**: new sources, measurements, or links can create valuable data
- **Collect systematically**: use the most structured route available and preserve raw files
- **Enable reproducibility**: script transformations, save definitions, and explore and check results
- **Handle big data**: Store in Parquet, query with DuckDB, and back up outside Git

Exercise: Discuss potential nonstandard data sources

- Split into groups of 2
- Each student presents one of their “ambitious” data sources
- The other student provides thoughts on how to make it feasible

For Monday October 5: Lightning Project Proposal

- Each student will have 5 minutes to present a single slide on their proposed project
- Highlight research question, proposed strategy to answer it, and why it matters
- Advice:
  - → Avoid overloading the slide with text
  - → Practice a timed presentation several times